11.9 Evaluating Numeric Expressions – Part 2

Lesson Introduction

Benchmarks	MA.7.NSO.2.1 Solve mathematical problems using multi-step order of operations with rational numbers including grouping symbols, whole numbers including grouping symbols, whole-number exponents, and absolute value. MA.7.NSO.2.2 Add, Subtract, multiply, and divide rational numbers with procedural fluency.	Learning Target	• I can evaluate numeric expressions with up to six operations.

Benchmark Coherence	Building On	Working Toward
	MA.6.AR.1.3 Evaluate algebraic expressions using substitution and order of operations. MA.6.NSO.1.4 Solve mathematical and real-world problems involving absolute value, including the comparison of absolute value.	MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.

Essential Questions	• How can I evaluate numeric expressions with up to six operations? • How can I apply what I've learned in this unit to evaluate expressions with up to six operations?
Lesson Narrative	Lesson 9 builds on the previous lesson by providing more practice and by incorporating more steps to evaluate each expression. Students will be asked to evaluate expressions that contain several operations and will require them to apply all that they have learned throughout this unit to complete the evaluations.

Component Summary				
	11.9.1 Warm-Up (~5 min)	Bell Work evaluating expressions with grouping symbols *(SE p. 206)*	11.9.4 Try It (10 min)	Using the order of operations activity *(SE p. 208)*
	11.9.2 Guided Instruction (5-10 min)	Evaluating expressions with six operations *(SE p. 206)*	11.9.5 Activity (10-15 min)	Class math debate activity *(SE p. 209)*
	11.9.3 Activity (10-15 min)	Dice game with evaluating expressions *(SE p. 207)*	11.9.6 Wrap-Up (~5 min)	Self-reflection on today's lesson *(SE p. 209)*

Resources	Glossary	Materials	Additional Preparation
	Key Terms: N/A Additional Terms: • Order of operations	• Dice (alternate option is a spinner) • (Optional) Paper to create labels for corners A, B, C, and D	• 11.9.5 Activity requires the corners of the room to be labeled A, B, C, or D. • (Optional) For 11.9.4 Try It, take the table and cut it into strips with the step numbers, the expressions, and a premade explanation.

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may misuse the order of operations by considering it a rule, not a guidance, that works with laws of exponents and properties of equality.
Students may make computation errors when evaluating.
Students may struggle with nested grouping symbols when there are several steps.</td><td>Consider preparing your groupings and having them sit in those seats prior to launching the lesson in order to reduce movement.
Students should not use a calculator for this lesson.
If time is an issue, then consider 11.9.4 Try It for homework.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Collect and Display
11.9.3 Activity has students roll dice to place values into a prewritten expression. Consider scribing what you see and hear to display for all students to view and use. This provides feedback for students in a way that increases accessibility while simultaneously supporting meta-awareness of language.</td><td>MA.K12.MTR.3.1: Mathematical Fluency
One of the primary goals of this lesson is for students to be able to apply what they’ve learned to evaluate numeric expressions. As such, students should be able to evaluate them with a high degree of fluency, since they have been working on this for two lessons. While the expressions may become lengthier, the process has remained the same.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">11.9.3 and 11.9.5 Activities have students working together with a partner or in small groups. Consider grouping students in a high/low performance grouping based on the data from the previous lesson. This can help struggling students learn from their peers while strengthening the other students’ understanding of the concept.

11.9.5 Activity incorporates a great deal of movement for students. Additionally, it provides an opportunity for students to verbalize their thinking around choosing the correct expression.

11.9.4 Try It allows students to try using the order of operations in an independent setting. It also has them identify and explain the steps. Student performance on question 1 can highlight any potential misconceptions that may need to be addressed with an individual student.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

11.9.1 Warm-Up: Bell Work

Component Overview: *Students will complete a Bell Work activity in order to explain the steps needed to evaluate the provided expressions. This activity is designed to be done independently.*

11.9.1 Warm-Up: Bell Work

Explain the steps needed to evaluate the expressions.

1. $(10 + 18 \cdot 2)^5 \div 3$
   Inside the parenthesis multiply 18 by 2, then add the product to 10. Take the value of that to the fifth power. Then, divide the result by 3.

2. $[(7 - 10)^2]^4 \times 2$
   Inside the parenthesis, subtract 10 from 7. Then, take the value of that to the second power. Then, take the result to the fourth power. Lastly, multiply the value by 2.

With this activity, notice that these questions are not asking for students to evaluate them but rather explain the steps needed to evaluate.
In question 2, students could also state that the difference of $7 - 10$ is taken to the power of 8.

Consider creating this as a partner discussion activity where they can describe question 1 with their partner and then switch for question 2. Engage students in a discussion of the results by having them share out their steps either with one student per problem or having each student share the next step in evaluating the expression.

11.9.2 Guided Instruction: Evaluating Expressions with Multiple Steps

Component Overview: *Students will receive formal instruction on evaluating expressions when there are six operations involved by exploring steps used by two different students.*

11.9.2 Guided Instruction: Evaluating Expressions with Multiple Steps

Two students evaluated the expression below in two different ways. Review the work of each student. Then, answer the questions.

Diego	Valeria
$[0.5 \times (0.5 + 3.5)^2]^3 + 26 \div 13$	$[0.5 \times (0.5 + 3.5)^2]^3 + 26 \div 13$
$[0.5 \times 4^2]^3 + 26 \div 13$	$0.5^3 \times (0.5 + 3.5)^{2\times3} + 26 \div 13$
$[0.5 \times 16]^3 + 26 \div 13$	$0.5^3 \times (0.5 + 3.5)^6 + 26 \div 13$
$8^3 + 26 \div 13$	$0.5^3 \times 4^6 + 26 \div 13$
$512 + 26 \div 13$	$0.125 \times 4^6 + 26 \div 13$
$512 + 2$	$0.125 \times 4{,}096 + 26 \div 13$
514	$512 + 26 \div 13$
	$512 + 2$
	514

1. How were Diego's steps and Valeria's steps different from each other?
   Answers will vary. Diego used order of operations to simplify the expression whereas Valeria used the laws of exponents.

2. Discuss with your partner whether you think one method was easier to use to evaluate the expression over the other.
   Answers will vary. Order of operations is easier for me to visualize the steps of the expression.

For this activity, consider beginning by having students work with a partner to review the work of each student. Have them discuss what they notice and what they wonder about each method.
Both Questions 1 and 2 can be done through a class discussion forum. Instead of having students write their answers, have them discuss with their partner and facilitate a whole-class review of each one.

- *Would you determine the answer using a different set of steps?*
- *Why do the nested grouping symbols make it possible to determine the value for $26 \div 13$ in the first step? (The division can be found because it would have to be done before the addition to the value that is produced from the nested grouping.)*

11.9.3 Activity: Expression Game

Component Overview: *Students will work in groups to complete a game where they roll dice to fill in the blanks an expression with the values provided by the dice. For this activity, students should work in small groups to complete.*

11.9.3 Activity: Expression Game

1. Each group will use a standard number cube with sides numbered $1-6$ to play a game. One player rolls the number cube. Each player decides which box to place that number in the expression. Continue rolling the cube until all blanks in the expression are filled. Evaluate your expression and record your score based on the point system in the table. If on any round two players' expressions have the same value, both players **lose** 10 **points.**

Answers will vary for this activity.

Round 1 **Scoring** – the player with the greatest value **earns** 8 points.	$[(\_\_\_\_ \times \_\_\_\_^2) + \_\_\_\_] \div \_\_\_\_$	My answer Points earned
Round 2 **Scoring** – the player with the least value **earns** 5.5 points.	$[(\_\_\_\_ \times \_\_\_\_)^2 + \_\_\_\_] \div \_\_\_\_$	My answer Points earned
Round 3 **Scoring** – the player with the greatest absolute value **loses** 3.7 points.	$(\_\_\_\_ \times \_\_\_\_^2) + (\_\_\_\_ \div \_\_\_\_)$	My answer Points earned
Round 4 **Scoring** – the player with the absolute value closest to zero **earns** 11 points.	$(\_\_\_\_ \times \_\_\_\_)^2 + (\_\_\_\_ \div \_\_\_\_)$	My answer Points earned

If no dice are available, consider using an online random number generator, spinners, or drawing precut numbers (one through six) from a plastic bag. The key feature of this activity is for the numbers to be random so that they can get additional practice with evaluating expressions using any value.

For a challenge, consider having precut rational numbers in a plastic bag. Have students make at least one or two of their values come from the bag instead of the whole numbers from the dice.

11.9.3 Activity: Expression Game (continued)

My Game Score

Round 1	Round 2	Round 3	Round 4	My Score

2. Discuss with the other players what strategies they used to determine where to put the numbers. Summarize your discussion.
Answers will vary based on the numbers rolled on the dice by each student.

As a way to close out this activity, allow select students to share out their answer to the last question. This can help settle the class back down and tie back in the mathematics prior to moving onto the next activity.

11.9.4 Try It: Evaluating Expressions with Multiple Steps

Component Overview: *Students will practice using the order of operations in order to evaluate expressions, including the steps and the explanation of each step. For this activity, students should work independently to practice what they have been learning over the past two lessons.*

11.9.4 Try It: Evaluating Expressions with Multiple Steps

1. In the table an expression has been evaluated, but the steps are mixed up. Order the expressions using the numbers 1 – 7 on the left. Write an explanation for each step.

Step	Expression	Explain the Step
3	$\lvert -11.591\rvert + 15 - (-3.1)^2$	*Evaluate inside the absolute value grouping symbol*
5	$11.591 + 15 - 9.61$	*Evaluate the exponent*
4	$11.591 + 15 - (-3.1)^2$	*Find the absolute value*
1	$\lvert 18.2 + (-3.1)^3\rvert + 15 - (-3.1)^2$	Original expression
6	$26.591 - 9.61$	*Evaluate from left to right- addition*
2	$\lvert 18.2 + (-29.791)\rvert + 15 - (-3.1)^2$	*Evaluate the exponent in the absolute value grouping*
7	16.981	*Subtract the values for final answer*

Consider taking this table and cutting it into strips with the step numbers, the expressions, and a premade explanation. Allow students to work in partner pairs to sort them out and try to match them in the correct order for the table.

The expression $\lvert 18.2 + (-3.1)^3\rvert + 15 - (-3.1)^2$ *can be evaluated differently than using the steps shown. For example, students who have a grasp of the order of operations, properties of equality, and laws of exponents may prefer to use these steps:*

$$\lvert 18.2 + \ -29.791\rvert + 15 - 9.61$$
$$\lvert -11.591\rvert + 5.39$$
$$11.591 + 5.39$$
$$16.981$$

The intent is not for students to be forced to do one step at a time when evaluating expressions but to ensure they can explain the steps. Alternatively, your students could show their own work and explain the mathematics that were applied.

11.9.4 Try It: Evaluating Expressions with Multiple Steps (continued)

2. Evaluate each expression without using a calculator.

   a. $\left(\frac{3}{4}\right)^2 \times 32 + 3^2$

   $\frac{3^2}{4^2} \times 32 + 3^2$

   $\frac{9}{16} \times 32 + 9$

   $18 + 9$

   27

   When monitoring student performance on question 2, watch for their steps more than the final answer. It is probable that students may make a calculation error without their calculator, but it is important that their process be correct.

   b. $[(10 + 4.5)^2 - 10.25]^2 \div 40$

   $[(14.5)^2 - 10.25]^2 \div 40$

   $[210.25 - 10.25]^2 \div 40$

   $200^2 \div 40$

   $40000 \div 40$

   1000

   c. $6.7 - (2^3 - 3)^3 \div 10 + 4$

   $6.7 - (8 - 3)^3 \div 10 + 4$

   $6.7 - 5^3 \div 10 + 4$

   $6.7 - 125 \div 10 + 4$

   $6.7 - 12.5 + 4$

   $-5.8 + 4$

   -1.8

11.9.5 Activity: Class Debate

Component Overview: *Students will engage in a class debate over choosing an expression that will result in a specific value. This activity is for the entire class but initially done independently.*

11.9.5 Activity: Class Debate

James wants to choose an expression that has a value of -778. He can choose between the expressions shown.

Expression A	Expression B
$14-(10\times 3^2)^2+6\times 3$	$14-10\times (3^2)^2+6\times 3$
Expression C	**Expression D**
$(14-10)\times (3^2+6)^2\times 3$	$14-[10\times (3^2)^2]+6\times 3$

Which expression should James choose?
Expressions B and D are equivalent to -778.

- Each corner of the room is labeled A, B, C, or D. After selecting the expression that you think James should choose, go to the corner of the room that matches the label for that expression.
- With your classmates who selected the same expression, discuss why your expression is the expression that James should choose. Discuss why James should not choose the other expressions. One member of your group will act as a representative during a class debate.
  Answers will vary.
- For the debate, prepare an opening statement and potential rebuttals for arguments from other groups. Keep arguments and rebuttals centered on the mathematics.
  Answers will vary.
- After the debate, explain whether you would select the same expression again.
  Answers will vary.

Prior to beginning this activity, consider reviewing the instructions with your students before releasing them to complete.

Students should be instructed to try and determine their own answer within a given time frame. Then, once time is called, they can complete the remaining parts of the activity that are listed in the boxes.

Consider adding an additional layer of engagement. Allow the group members to move to a different corner (if they think they should) after each presentation to see if anyone has convinced others to join their expression.

11.9.6 Wrap-Up: Ticket Out the Door

Component Overview: *Students will complete a ticket out the door on their learning from this lesson. This activity should be done independently to understand each student's level of proficiency.*

11.9.6 Wrap-Up: Ticket Out the Door

1. Evaluate each expression without a calculator. Show each step you use.

$$-\left|5-\frac{3}{5}\right|+\frac{4}{15}(4^2-61)$$

Work

$$=-\left|\frac{22}{5}\right|+\frac{4}{15}(4^2-61)$$
$$=-\frac{22}{5}+\frac{4}{15}(4^2-61)$$
$$=-\frac{22}{5}+\frac{4}{15}(16-61)$$
$$=-\frac{22}{5}+\frac{4}{15}(-45)$$
$$=-\frac{22}{5}+-12$$
$$=-\frac{82}{5}$$

$$\frac{(1.5+|-3.25-4.25|)^3}{12}$$

Work

$$=\frac{(1.5+|-7.5|)^3}{12}$$
$$=\frac{(1.5+7.5)^3}{12}$$
$$=\frac{(9)^3}{12}$$
$$=\frac{729}{12}$$
$$=\frac{729}{12} \text{ or } =\frac{243}{4}$$

Proficiency on this Wrap-Up includes:

- *Understanding which operations should be done first. Students may choose to perform more than one operation first by considering the properties of equality.*
- *Understanding that one value should be determined from an expression inside an absolute value or a set of parentheses.*
- *Fluency of operations with rational numbers. A small miscalculation that a student carries on throughout their work should be considered a minor mathematical error. Fluency can be shown despite a small miscalculation. Teachers are encouraged to consider a student's work when determining fluency.*